Executive Business & Technical Report

Project Title: Unlocking Behavioral Intelligence in Airline Loyalty Programs

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Title Page & Executive Summary

1.1 Project Overview

Modern airline loyalty programs generate massive transaction data, but many remain cost centers that focus on points accumulation and rewards rather than serving as proactive engines for customer retention. When an airline waits for a customer to formally cancel their membership, the intervention is usually too late. The customer has already built habits with a competitor, and winning them back is extremely expensive.

This report presents a **proactive behavioral intelligence system** designed for **FlyCaelum Airlines**. By leveraging monthly flight transactional records and customer profile histories of approximately 16,700 Canadian members from 2012–2018, we build a predictive machine learning pipeline that flags "silent churn" (behavioral disengagement) before formal cancellation occurs.

To achieve this, we introduce the **Behavioral Disengagement Score (BDS)**—a composite behavioral metric that quantifies customer disengagement based on flight inactivity, points velocity, and redemption dormancy. Using this score, we define two classes of churn: **Soft Churn** (members who have behaviorally disengaged but not formally cancelled) and **Hard Churn** (members who have formally cancelled their account). By training predictive models on this behavior-grounded target variable, we identify disengaged members with high accuracy.

We then integrate these risk scores with Customer Lifetime Value (CLV) to segment our members into six actionable personas—including a newly identified **"Ghost Members"** category—and map them to targeted, trigger-based retention campaigns.

1.2 Key Outcomes & Metrics

- **Temporal Validation Integrity**: Implemented a snapshot cohort design (t0 = 2018-06-01 snapshot) for all model training and testing. This ensures the model is evaluated on future, unseen cohorts, mimicking a true production deployment and guaranteeing evaluation integrity.
- **Model Performance**: The time-aware **Random Forest** and **XGBoost** models achieved strong predictive power with test cohort **ROC-AUC of 0.938** and **0.942**, respectively.
- **High-Precision Risk Scoring**: At the default classification threshold, the XGBoost model achieved **91.4% Precision** for predicting disengagement, meaning that out of 100 flagged members, 91 are true churners, minimizing wasted retention spending.
- **Customer Value Segmentation**: A KMeans segmentation clustered active members into behavioral personas (Champions, Corporate Flyers, Active Leisure) based on CLV, salary, and predicted risk, calibrated with rule-based overrides for **New Members (Onboarding)** and **Ghost Members**.
- **CLV Trajectory Insight**: Average CLV is almost identical between active ($7,957.22 CAD) and disengaged ($8,039.21 CAD) customers. This proves that historical value accumulation (CLV) is not a predictive signal of future engagement, validating our focus on behavioral trajectories.

- **Advanced Survival Insights**: Fitting a Cox Proportional Hazards model revealed that each month of program tenure decreases churn hazard by **3%** (Hazard Ratio = 0.97, $p < 0.005$), while each month of flight inactivity increases churn hazard by **12%** (Hazard Ratio = 1.12, $p < 0.005$).
- **Reactivation ROI**: Successfully reactivating just **15%** of the high-risk, high-value *At-Risk Elites* segment in a single cohort recovers **$95,931.80 CAD** in customer lifetime value. After accounting for campaign costs, this yields a **net business value saved of $86,631.80 CAD** with an estimated **ROI of 931.5%**.
- **Ghost Member Campaign Savings**: Excluded **253 Ghost Members** from campaign lists. Assuming a $100 campaign cost per head, this directly saves **$25,300.00 CAD** in marketing waste and prevents lists pollution.

Problem Framing & Business Context

2.1 The Silent Churn Challenge

Airlines rely heavily on repeat business. A small percentage of highly active customers often generates a massive portion of total passenger revenue. However, traditional loyalty programs fail to identify disengagement in a timely manner. Churn in airline loyalty programs manifests in two ways: 1. **Hard Churn (Explicit Cancellation)**: The customer formally closes their account. This is rare and usually happens long after the customer has stopped flying. 2. **Soft Churn (Silent Churn)**: The customer remains enrolled in the loyalty program but stops booking flights, stops redeeming points, and begins flying with competitor airlines.

Soft churn is highly dangerous because it is invisible to standard transactional reports. Identifying the exact point at which a customer transitions from "temporarily inactive" to "permanently disengaged" is the core challenge.

2.2 The Behavioral Disengagement Score (BDS)

To quantify soft churn, we engineer a composite metric, the **Behavioral Disengagement Score (BDS)**. BDS aggregates three key behavioral indicators: * **Flight Inactivity (40% weight)**: `Recency_Flights / 24`, clipped to `[0,1]`, reflecting the time elapsed since the last flight relative to a two-year window. * **Redemption Dormancy (30% weight)**: `Recency_Redemptions / 24`, clipped to `[0,1]`, reflecting points redemption inactivity. * **Point Momentum (30% weight)**: `1.0 - Points_Velocity`, clipped to `[0,1]`, measuring the change in points accumulation frequency in the last 3 months relative to the last 12 months.

The BDS formula is defined as:

BDS Score = (0.40 × Flight Inactivity) + (0.30 × Redemption Dormancy) + (0.30 × Point Momentum)

A customer is classified as in **Soft Churn** if their BDS exceeds **0.55** without formal cancellation. The final model target label, **Churn_Target**, is defined as 1 if the member is in Soft Churn OR has a non-null cancellation date (Hard Churn), and 0 otherwise.

Dataset Overview & Data Quality Assessment

3.1 Dataset Profiling

The analysis is conducted on three primary raw datasets: * `Customer Flight Activity.csv` (392,936 rows): Monthly transactional logs recording flights booked, distance traveled, points accumulated, points redeemed, and the dollar cost of redemptions. * `Customer Loyalty History.csv` (16,737 rows): Static demographic profiles containing country, province, city, gender, education level, annual salary, marital status, loyalty card tier (Star, Nova, Aurora), lifetime CLV, and enrollment/cancellation dates. * `Calendar.csv` (2,557 rows): Temporal lookup tables.

3.2 Advanced Data Cleaning & Imputation

During data profiling, several critical data quality steps were implemented to prepare the cohort: * **Negative Salary Correction**: Negative values in the `Salary` column were converted to positive values via absolute transformation (`.abs()`). Statistical analysis confirmed that the positive and negative salary ranges shared overlapping means and variances, verifying that these records represented sign-entry errors. * **Salary Imputation**: Missing values in the annual salary records were imputed using a conditional group median based on the customer's highest completed `Education` level and `Loyalty Card` tier, preserving demographic variations. * **Date Normalization**: Enrollment and cancellation years/months were combined into standard datetime dates to support precise time-difference features.

Machine Learning Pipeline & Temporal Cohort Design

4.1 Temporal Generalization and Evaluation Integrity

When designing predictive models for customer churn in contractless settings, traditional cross-validation (e.g., random K-fold splits) can lead to overly optimistic performance estimates. This occurs because customer transactional habits are time-dependent, and predicting future churn using past behaviors requires a clean separation of temporal windows. If features and target labels overlap in calendar time, the model can inadvertently learn lookup rules rather than true predictive patterns.

To ensure the model generalizes to future periods and simulates a true production deployment, we avoid time-overlapping features and construct a rigorous, time-aware cohort design.

4.2 Cohort-Based Temporal Validation Framework

To evaluate model performance, we implement a snapshot cohort design: * We select a snapshot date t0 (e.g., `2018-06-01`). * We identify the active cohort: members who enrolled before t0 and have not cancelled prior to t0. * **Features (X)** are engineered using transactional history strictly *prior* to t0 (e.g. up to May 31, 2018). * **Target Label (y)** is defined by looking *forward* into the target window [t0, t0 + 6 months] (June 1, 2018, to November 30, 2018) and checking for Soft Churn (BDS > 0.55) or Hard Churn. * **Temporal Splits**: To train and evaluate the model, we use two separate cohorts: * **Train Cohort**: Snapshot t0 = 2018-01-01. Features engineered on data prior to 2018; target observed in Jan-Jun 2018. * **Test Cohort**: Snapshot t0 = 2018-06-01. Features engineered on data prior to June 2018; target observed in Jun-Nov 2018.

Model Performance & Churn Risk Drivers

5.1 Machine Learning Results

We evaluated three classification algorithms on our temporal test cohort (13,473 active members). Because the base rate of churn in the target window is imbalanced (5.28%), we prioritize ROC-AUC and Precision-Recall AUC (PR-AUC):

Model	ROC-AUC	PR-AUC	Precision	Recall	F1-Score	F2-Score
Logistic Regression	0.8821	0.8021	98.4%	60.3%	74.8%	65.4%
Random Forest	0.9378	0.8664	93.1%	75.2%	**83.2%**	78.2%
XGBoost	**0.9422**	**0.8666**	**91.4%**	**76.0%**	83.0%	**78.7%**

The Random Forest and XGBoost models demonstrate excellent predictive power on the temporal test cohort. At a default threshold of 0.50, XGBoost achieves a **Precision of 91.4%** and **Recall of 76.0%**, meaning that the model is extremely reliable, preventing wasted marketing spend on false alarms.

5.2 Top Churn Risk Drivers

XGBoost feature importance (relative gain) reveals the top drivers of customer disengagement: 1. **Points Velocity (`Points_Velocity` - 38.6%)**: The change in points accumulation frequency is the single most important churn driver, reflecting shifts in booking behavior. 2. **Redemption Ratio (`Redemption_Ratio` - 30.2%)**: Customers who accumulate but never redeem points, or who suddenly redeem all points (draining their balance), represent different churn risks. 3. **Recent Points Accumulation (`Points_Accum_Last_3M` - 6.2%)**: A drop in points accumulated in the last 3 months indicates a decline in transaction frequency. 4. **Customer Tenure (`Tenure_Months` - 4.0%)**: Program tenure plays a key role. Long-term members who stop flying are high churn risks, while new members who haven't flown yet are just starting their customer lifecycle.

Strategic Customer Segmentation & Personas

Rather than clustering on raw mathematical variables alone, we built a **Value-vs-Risk Segmentation Matrix** using KMeans clustering on five key features: CLV, Salary, Flights_Last_12M, Recency_Flights, and Churn_Probability. We then overlay rule-based logic to handle onboarding and inactive accounts, resulting in six distinct customer personas:

Segment Profiles Summary Table

Persona Name	Avg. CLV (CAD)	Avg. Salary (CAD)	Annual Flights	Recency (Flights)	Churn Risk	Count
Loyal Champions	$9,659.83	$73,151.78	22.42	1.4 months	1.8%	6,284
Active Leisure	$6,270.49	$71,180.22	12.40	1.8 months	3.9%	6,187
Corporate Flyers	$7,510.68	$221,740.42	17.93	2.2 months	3.0%	352
New Members (Onboarding)	$7,910.66	$70,141.76	0.00	24.0 months	2.7%	304
Ghost Members	$8,205.12	$69,183.12	0.00	24.0 months	99.0%	253
At-Risk Elites	$6,876.83	$74,233.66	0.19	23.0 months	99.0%	93

The Value of Segment Calibration

A key demonstration of the model's intelligence is the difference between **At-Risk Elites** and **New Members**: * Both segments show high recency metrics (~23-24 months of no flights). * However, the model assigns **99.0% Churn Probability** to At-Risk Elites because they have a mature tenure with the program and have stopped flying (true disengagement). * It assigns only **2.7% Churn Probability** to New Members because they are brand new signups (tenure <= 2 months) who simply haven't booked their first flight yet.

CLV Comparison: Active vs. Churn Cohorts

To investigate whether historical value accumulation (CLV) prevents churn, we compared the average CLV between active and disengaged members in the test cohort:

Cohort Status	Average CLV (CAD)
Active / Engaged (Churn = 0)	$7,957.22
Churned / Disengaged (Churn = 1)	$8,039.21

As shown, the average CLV is nearly identical between the active and disengaged cohorts. This provides a crucial strategic insight: **historical value accumulation (CLV) is not a predictive signal of future engagement.** Instead, disengagement is driven entirely by the customer's behavioral trajectory, validating our focus on the time-aware BDS metric.

Advanced Loyalty Survival Analysis

To understand the temporal dynamics of customer retention and measure the hazard ratios of program tenure, we conduct a survival analysis using Kaplan-Meier curves and Cox Proportional Hazards regression.

6.1 Kaplan-Meier Retention Stratification

The Kaplan-Meier survival curves estimate the probability that a customer remains engaged (does not churn) over time. Stratifying these curves by loyalty card tier (Star, Nova, Aurora) reveals the retention probability at key milestones:

Program Tenure	Aurora Tier	Nova Tier	Star Tier
12 Months	94.57%	95.12%	95.04%
24 Months	93.25%	93.72%	93.65%
36 Months	91.82%	92.23%	92.30%
48 Months	89.96%	90.22%	90.18%
60 Months	87.77%	87.34%	87.58%
72 Months	**58.07%**	**67.12%**	**65.37%**

Key Finding: Retention remains high and stable across all tiers up to 60 months (above 87%). However, at **72 months**, the retention probability drops significantly for all tiers, with Aurora tier dropping most severely to **58.07%** (compared to 67.12% for Nova and 65.37% for Star). This suggests that high-tier benefits alone are insufficient to prevent disengagement if travel habits shift, making proactive intervention critical before the 5-year milestone.

6.2 Cox Proportional Hazards Model

We fit a Cox Proportional Hazards regression model to measure the effect of demographic and behavioral covariates on the hazard of churn. L2 regularization (`penalizer=0.1`) was applied to ensure model stability and convergence:

- **Program Tenure**: Each additional month of tenure is associated with a **3% decrease** in churn hazard (Hazard Ratio = 0.97, $z = -33.81$, $p < 0.005$). This proves that mature loyalty relationships are statistically more resilient.

- **Recency of Flights**: Each month since the last flight increases the hazard of churn by **12%** (Hazard Ratio = 1.12, $z = 46.98$, $p < 0.005$), making it the single most critical behavioral risk driver.
- **CLV and Card Tier**: Customer Lifetime Value ($p = 0.65$) and Card Tier ($p = 0.87$) are not statistically significant predictors of survival when controlling for behavioral variables. This confirms that disengagement is behaviorally driven rather than value-driven.

Smart Retention Framework & ROI Analysis

We map our behavioral segments directly to targeted, trigger-based retention campaigns:

7.1 Retention Campaign Playbook

At-Risk Elites → Priority Reactivation Concierge & Status Extension

- *Trigger*: Churn Probability > 0.50, Persona is "At-Risk Elites".
- *Action*: Direct phone outreach from premium service, 1-year elite tier status extension, and a high-value flight voucher ($150 CAD) for their favorite historical route.

Loyal Champions → Exclusive VIP Benefits

- *Trigger*: Persona is "Loyal Champions".
- *Action*: Airport lounge passes, priority boarding, and early access to promotional fares to maintain brand loyalty.

New Members (Onboarding) → Welcome Offer & First-Flight Voucher

- *Trigger*: Persona is "New Members (Onboarding)".
- *Action*: Automated onboarding emails offering a first-flight discount ($30 CAD) and 1,000 loyalty points upon their first booking.

Corporate Flyers → Business Perks & Upgrades

- *Trigger*: Persona is "Corporate Flyers".
- *Action*: Double miles promotions on corporate commuter routes, business class upgrade certificates, and co-branded credit card point matching.

Active Leisure → Seasonal Destinations & Partner Deals

- *Trigger*: Persona is "Active Leisure".
- *Action*: Early-bird vacation package discounts, discount point redemptions for resort destinations, and hotel/car rental partner point multipliers.

Ghost Members → Remove from Marketing Lists (Campaign Exclusion)

- *Trigger*: Persona is "Ghost Members".
- *Action*: Suppress from active campaign lists. This prevents budget waste on inactive accounts.

7.2 Business Impact & ROI Case Study

Let's analyze the financial ROI of targeting the **At-Risk Elites** segment (Segment 0) in our test cohort:

- **NET BUSINESS SAVINGS SAVED**: **$86,631.80 CAD** (Headline Net Savings)
- **Estimated Return on Investment (ROI)**: **931.5%** (Supporting ROI Metric)
- **Target Cohort Size**: 93 members.
- **Average Customer Lifetime Value (CLV)**: $6,876.83 CAD.
- **Total Portfolio Revenue at Risk**: $639,545.33 CAD.
- **Estimated Reactivation Success Rate**: 15% (Conservative industry baseline).
- **Average Campaign Cost (Offer Cost)**: $100 CAD per member (voucher + admin).
- **Total Campaign Cost**: 93 x $100 = $9,300.00 CAD.
- **Gross Revenue Saved**: 93 x 15% x $6,876.83 = $95,931.80 CAD.
- *Calculation*: Net Savings = $95,931.80 - $9,300.00 = $86,631.80 CAD. ROI = ($86,631.80 / $9,300.00) x 100 = 931.5%.

This highlights that targeting high-value slipping members is highly lucrative compared to broad-based promotions.

Limitations, Safeguards & Future Horizon

8.1 Project Limitations

- **Data Completeness**: The dataset records monthly flight transactions but lacks critical features like customer service complaints, website search history, competitor pricing, and check-in delays. These features are strong leading indicators of churn.
- **Macroeconomic Trends**: The data spans 2012–2018. Changes in macroeconomic factors, competitor routes, or airline alliances during this period are not modeled, which could impact the baseline churn rate.
- **Placeholder Default Recency**: Brand new signups are assigned a default recency of 24 months, which requires the model to rely heavily on tenure to distinguish them from churned members.

8.2 Operational Safeguards: Ghost Member Exclusion

A key financial safeguard of this system is the automatic identification of **Ghost Members** (accounts with tenure > 6 months, zero net points balance, and zero flights in the last 12 months). By suppressing these **253 members** from marketing lists, the airline immediately saves:

Direct Marketing Savings = 253 members × $100 campaign cost = $25,300.00 CAD

This direct cost reduction is achieved with zero impact on passenger revenue, as these accounts are functionally dead, showing the power of combining business-rule filters with machine learning pipelines.